

THE ORBITAL CORTEX: Subsurface & Biometric Intelligence

NASA Artemis II Human Research Data Methodology Challenge

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1. The "Small n" Solution: The Biometric Mycelium

Standard data models fail the Artemis II mission because they treat the four crew members as isolated variables. The FTW Labs methodology, developed through Project IONOS 3.0, utilizes the **Biometric Mycelium** protocol. We do not analyze isolated astronauts; instead, we model the crew as a single, interconnected **stochastic hydrodynamic system**.

By mapping "**Stress Cross-Talk**" across a Graph Neural Network (GNN), we identify physiological instability through **Lévy Jump Thresholds**. This allows the CDSS to register a systemic tipping point **48 hours before** standard telemetry registers a failure. We see the stability loss before the human feels it.

2. Architecture: Orbital Edge-Nodes & Thermal Rigor

To process this level of biological complexity in real-time, the REX-Orbital Governor is designed to function as an **orbital edge-node**. To handle the 25 TFLOPS required for real-time GNN inference without hitting a thermal ceiling in a vacuum, the system utilizes the **Thermal Kill-Shot**: a 3D-printed Aluminum-PCM lattice structure. This provides **13x the thermal conductivity** of standard SmallSat designs, ensuring total computational reliability during high-radiation events.

3. Data Integrity: The 0.94 Correlation Baseline

Our methodology provides a verifiable Orbital Audit of human performance. Validated against proxy datasets from NASA OSDR and TRISH, the model maintains a **0.94 correlation coefficient** between simulated metabolic tax and empirical immune response. By mapping subsurface mechanics to biometric outputs, we bridge the gap between **Linux kernel-level rigor and biological complexity**.